

The spatial integration and coordinated industrial development of urban agglomerations in the Yangtze River Economic Belt, China

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ABSTRACT

Urban agglomeration is the engine of national development and regional prosperity. Although extensive work has investigated issues related to this new form of spatial governance, few studies have directly illustrated the spatial integration of urban agglomeration and its relationship with industrial development. This paper employs nighttime light data and industrial enterprise datasets to investigate the spatial integration and industrial development in the Yangtze River Economic Belt (YREB) of China for 1995–2015. We here illustrate the significant relationship between the spatial integration of urban agglomerations and the characteristics of industrial development. In the process of spatial integration, urban form, intercity relation and their evolution show clear regional differences. Because of the differences in socio-economic and geographical characteristics, urban systems are more advanced and closely related in developed areas. A significant negative (positive) spatial correlation between industrial specialization (diversification) and urban form is supported by using bivariate Moran's I, and spatial clustering patterns are clearly different across the three urban agglomerations. A panel regression reveals that intercity relations are significantly associated with the characteristics of industrial development. Higher levels of industrial diversification and competition are associated with weaker intercity relations, while industrial structures similarities are reversed. These findings could be used to formulate reasonable policies and plans and to support future regional spatial integration and coordinated development.

1. Introduction

Urban agglomeration is the production of rapid globalization, urbanization and spatial reconstruction and has gradually become a new form of spatial governance and a driving force behind economic growth. Governments and societies are actively embracing this transformation, and a series of development plans has been proposed to envision urban agglomeration in the Global North and Global South. For example, The Regional Plan Association has proposed the “American 2050 Plan” for 11 Megaregions. The central government of China has also formulated three development plans for the Beijing-Tian-Hebei Urban Agglomeration, the Yangtze River Delta Urban Agglomeration and the Guangdong-Hong Kong-Macau Big Bay Area to drive economic growth and regional development. Urban agglomerations stem from the continuous expansion of urban or metropolitan areas and from the blurring of their boundaries where interlocking economic systems and shared natural resources and ecosystems are

connected by various transportation networks to reallocate labour and capital at a low cost and over a short period of time (Florida, Gulden, & Mellander, 2008). The characteristics of urban agglomeration determine that its evolution is not immune to the effects of industrial differences, local competition and government intervention. Associated formation and evolution processes and their determinants are not homogenous across different regions and countries. Further, differences in development goals, agglomeration capability and local advantages also likely lead to spatial inequality and sub-optimal structures of urban agglomeration, repressing growth potential and obstructing the sustainable development of urban agglomerations (Wei, 2015). It is thus urgent that sufficient evidence and guidance for regional spatial integration and coordinated development should be identified. Meanwhile, scholars advocated numerous studies on urban agglomeration issues from space-time form-dominant or function-dominant perspectives, making it beneficial to determine the relationship between research paradigms of different schools and to identify avenues for future

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urban agglomeration research (Harrison & Hoyler, 2015). With the rise of protectionism and de-globalization, reexamining the relations and linkages between local cities also has important implications for urban and regional development in the current era.

Considering the prominent importance of urban agglomerations and the increasingly complex internal and external situations, the authorities urgently need reasonable planning strategies to achieve the coordinated development of urban agglomerations. In this context, describing the spatial integration within urban agglomerations and examining the relationship between spatial integration and industrial development are extremely important. Although the urban agglomeration has attracted extensive research focused on its definition (Fang & Yu, 2017), spatial structures (Gao, Huang, He, & Dou, 2017; Zhen, Cao, Qin, & Wang, 2017), development mechanisms (Liu, Zhang, Kong, Wang, & Chen, 2018; Shao, Tian, & Yang, 2017; Ye, Zhu, Li, Yang, & Chen, 2019) and other aspects, few studies have addressed these two issues directly. First, urban systems are studied by using intercity flow data from a network perspective. Although methods and studies of network paradigms are effective at revealing processes of urban system formulation at the global scale (Meijers, 2007), they ignore the directivity of intercity relation and cannot measure the differences of two directions and their spatial evolution. They are also limited to reflect the effects of local factors such as development strategies, government intervention, and cultural identity. Furthermore, governance or plan demands of a visible morphological space cannot be overlooked. Second, throughout urban and regional development, industries constantly develop, transform and upgrade (Martin, 2010; Martin & Sunley, 2006). The relationships between spatial integration and characteristics of industrial development (e.g., specialization, diversification and competition) and how such factors affect the formation and development of urban agglomerations remain ambiguous. Third, mechanisms of urban agglomeration development in China differ from those of centrally orchestrated city-regionalism observed in the UK (Harrison, 2008) and of city-regionalism observed in the US (Brenner, 2002; Jonas & Pincetl, 2006), which can be treated as a combination of the two modes (Li & Wu, 2017). Such a feature could provide more nuanced understandings for the spatial integration and industrial development of urban agglomerations. Therefore, the present study aims to fill these gaps by addressing the two following questions:

- (1) How is the spatial integration within the urban agglomeration conducted, and how is it measured?
- (2) What role do industrial development play in spatial integration?

The rest of this paper is organized as follows. The following section synthesizes the literature on the spatial integration of urban agglomeration and its relationship with characteristics of industrial development. After introducing our selected data and methodological design, the fourth section illustrates the results of our empirical analysis on the process of spatial integration and the relationship between it and industrial development. The fifth section discusses the implications for policy and planning of spatial integration and coordinated development in China. The final section concludes the paper and lists potential biases and avenues for future research.

2. Literature review

2.1. Emergence of urban agglomerations: spatial integration and its measures

Space is a source of economic (dis)advantages, geographical advantages and advantages springing from the cumulative nature of productive processes in space. It determines the endowments of production factors, the accessibility of an area and the production and transition costs (Capello, 2011). The spatial structure and form of urban spaces and urban agglomerations, as spatial entities with high

concentrations of population and socio-economic activities, are constantly changing. In this transformation and reconstruction process, the development of urban and regional areas is affected by this series of spatial advantages and limitations. Their development is also a path dependence process, which is an endogenous and self-reinforcing process shaped by its historical trajectory and by previous competencies such as technologies, institutions, labour skills and industrial development (Martin, 2010; Martin & Sunley, 2006). In the polis epoch, the agglomeration of non-agricultural populations could be maintained due to a surplus of agricultural production (Jacobs, 1969). Although government administration, religious sacrifice and handicraft production are found in cities, the number of cities and their scale are still limited. Cities are then dispersed in space with weak connectivity, and most have a mononuclear structure from their inception. Based on neo-classical demand and substitution theory, Alonso and his successors extended the 'Thünen Kreise' agricultural rent theory to develop monocentric city models that assume that goods production occurs in only one central contiguous area (Alonso, 1964; Mills, 1967; Muth, 1969). However, this model is not suitable to explain several cities or urban systems because it reaches the somewhat paradoxical conclusion that those cities must all be of the same size (Capello, 2011). In subsequent development and related research, market supplies, transportation costs, production profits, resource acquisition and other factors were further stressed in location theories and were considered during site selection (McCann & Sheppard, 2003). After the second Industrial Revolution, to take advantage of strategic location and infrastructure, urban spatial forms and structures began developing in a non-concentric manner, with spatial structures gradually becoming more complex. According to Morrill (2006), these underlying city cores scattered established and mushroomed in space and experienced the following processes: (1) sheer economic and demographic growth, (2) physical decentralization in the form of suburbanization, (3) extension of metropolitan commuting fields and the physical coalescence of formerly physically separate areas, (4) rise or restructuring of and reaching out to formerly distant satellites, and (5) restructuring and revitalization of high-level metropolitan cores. Meanwhile, negative externalities of urban expansion and the monocentric development of regions and cities (e.g., diseconomies of scale, traffic congestion and air pollution) also promoted these processes (Agyemang, Amedzro, & Silva, 2017; Huang, Liu, Zhao, & Zhao, 2017; Tan, 2017). Entering the early 1970s, dominant industries of the social economy were no longer confined to manufacturing, and the rise of high-tech, service, design, cultural and other industries transformed traditional manufacturing from the Global North to the Global South. A new 'post-Fordist' economy started to prevail and network cities with goals for economic efficiency gradually emerged (Scott & Storper, 2015). This transformation led to urban development across the globe and furthered the diversification of commodity supplies and consumption and levels of local specialization (Cao, Shi, & Liu, 2016). Potential for the rapid development of cities of varying ranks emerged, providing a basis for the emergence of urban agglomerations. Adjacent metropolitan areas, cities and other settlements are now bound together through expressways, high-speed railways, beltways and other transportation infrastructures, which form urban agglomerations (Lang & Knox, 2009).

As a representation of urban and regional development, the evolution of spatial structures and forms can provide information on spatial reconstruction and integration. Urban space form could be measured by spatial metrics from five dimensions: compactness, centrality, complexity, porosity and density (Huang, Lu, & Sellers, 2007). Prior studies have employed remote sensing data of varying resolutions (Chen, Liu, & Li, 2017; Jiao, 2015; Ma et al., 2014; Wei, Li, & Yue, 2017; Xu, He, Liu, & Dou, 2016; You, 2016; Zhang et al., 2016) or socio-economic spatial datasets and census data (Zhang, Sun, & Li, 2017) to deliberate on spatial structures, spatial forms and the distributions of populations and economic activities. However, a single data source can only reflect the narrow concept of urban form and structure, and some statistical

datasets cannot measure spatial changes occurring in a city or region. Nighttime light data, as a reliable proxy for urban development and change, not only reflects the physical properties based on land use changes but also shows spatial changes of several socio-economic characteristics. Nighttime light data has the characteristics of continuity, credibility, accuracy, and comparability. It can also go beyond city boundaries to capture the spatial distribution of populations and their socio-economic activities within both urban and peri-urban regions (Liu, Zhang, et al., 2018). Therefore, nighttime light data could provide an effective way to explore traditional and novel issues by combining with other datasets and is playing an increasingly important role in urban and regional studies. In physical properties and morphological research, DMSP/OLS dataset is the most common nighttime light data because of its sufficient data storage. DMSP/OLS dataset is widely used to map urban extent (or boundary) (Hu, Qian, Pickett, & Zhou, 2020; Ju, Zhou, & He, 2016; Wang, Wan, Guo, Hu, & Zhou, 2017) and extract impervious surface (Cheng, Luo, Shi, Xia, & Shen, 2020) at various scales (Li & Zhou, 2017). It also provides an effective way to delineate the spatiotemporal characteristics and driver forces of urban expansion (or sprawl) (Duque, Lozano-Gracia, Patino, Restrepo, & Velasquez, 2019; Guan, He, He, Cheng, & Qu, 2020) and shrinkage (Zhou et al., 2019). With the improvements of the satellite sensors of nighttime remote sensing, NPP/VIIRS and Luojia1-01 provide finer spatial resolution datasets, which makes the application of nighttime light data continuously popularized in urban and regional studies (Yin, Li, Tong, Li, & Jendryke, 2020). Some scholars are starting using NPP/VIIRS and Luojia1-01 datasets to identify the urban centres and explore inner-city structure (Cai, Huang, & Song, 2017; Chen et al., 2017). Technical progress and research records have greatly promoted the studies of more complex and multi-dimensional urban systems, such as their spatial structure (Peng et al., 2020; Zhong, Lin, Zhou, & Chen, 2018), city-size distribution (Gao et al., 2017; Wu et al., 2019) and polycentricity (Cai et al., 2017). In socio-economic development and urbanization studies, scholars have confirmed that nighttime light data has significant relationships with some socio-economic factors such as economic and infrastructural growth (Keola, Andersson, & Hall, 2015; Nordhaus & Chen, 2015), population distributions (Anderson, Tuttle, Powell, & Sutton, 2010), energy consumption (Elvidge et al., 2009), and processes of urbanization (Gao, Huang, He, & Ma, 2015; Liu et al., 2015; Zhang & Seto, 2011). As the research progresses, nighttime light data also has been employed to study urban vitality (Guan et al., 2020; Xia, Yeh, & Zhang, 2020), spatial spillover effect (Zheng, Long, Chang, & Ye, 2019), ghost cities (Jin et al., 2017; Zheng et al., 2017), housing price (Li, Zou, Wu, & Xu, 2019) and other issues. Scholars have conducted a lot of substantial research on these two aspects, which has laid a solid foundation for future urban and regional studies. However, few studies have applied nighttime light data to directly study the spatial integration of urban agglomerations. The dual attributes of nighttime light data, including physical and socio-economic attributes, still need to be further explained and applied. In addition, the understanding and measurement of intercity relation could be divided into two categories (Meijers, 2007): (1) Based on the theory of central place, intercity relation could be understood as competitive and hierarchical. In relevant studies, the gravity model and its improved models have been used to quantify the intercity relation from this perspective (Krings, Calabrese, Ratti, & Blondel, 2009; Ye et al., 2019). (2) According to the flowing space theory, the intercity relation could be identified by the flow of labour, capital, information, etc. Many studies have employed an interlocking network model to study intercity relations and urban networks (Yeh, Yang, & Wang, 2015; Zhao, Derudder, & Huang, 2017). In this study, the intercity relationship aims to reflect the spatial tendency and preference of a city towards its geographical neighbours. However, these methods would only obtain a single value for a pair of cities, which means they ignore the directivity of intercity relations and cannot measure the differences of two directions and their spatial evolution. In other words, these methods cannot effectively reflect the

tendency of urban development in the spatial integration of urban agglomerations. Therefore, to precisely delineate the process of spatial integration, a new quantitative method needs to be constructed.

2.2. The spatial integration of urban agglomerations and industrial development

The formation of urban agglomerations is a process through which industries constantly develop, transform and upgrade. In addition to the impact of industrial vicissitudes on spatial forms and structures, the geographic concentration and agglomeration of industries also drives changes in production space (Guo & He, 2017) and then affects the spatial evolution of urban agglomeration together with physical characteristics. Scott and Storper (2015) argue that the formation and evolution of all cities can be understood through dynamic processes of agglomeration/polarization and from interactive processes of land use. The externalities that stem from spillover effects of industrial agglomeration are believed to contribute to city formation, growth and decline and can be divided into Marshallian and Jacobian externalities (Jacobs, 1969; Marshall, 1999). They promote the sharing (local interlinkages within production systems and the supply of public goods), matching (connections between local pools of firms and labourers) and learning (the flow of formal and informal information) of knowledge and technologies among enterprises through informal market mechanisms (Duranton & Puga, 2004; Scott & Storper, 2015). Among them, Marshallian externalities are expected to stimulate incremental innovation and spillover effects originating from similar firms producing similar products, and Jacobian externalities are expected to spur radical innovation and product innovation between different sectors and to forge new markets and employment needs (Frenken, Van Oort, & Verburg, 2007; van der Panne, 2004). It is no doubt that local systems, industry types and industrial networks will affect such externalities. Such differences in industrial development paths will also directly determine the evolution of spatial structures and intercity relations. Although a large body of work has examined the (dis)advantages of Marshallian and Jacobian externalities for regional development (Beaudry & Schiffauerova, 2009; Ercole & O'Neill, 2016; Feldman & Audretsch, 1999; Porter, 2000; van der Panne, 2004; Wei & Gu, 2013), research on regional spatial evolution and development in reference to industrial agglomerations must be furthered.

On the other hand, an extensive and rigorous body of literature has explored and demonstrated how spatial structures or forms affect industrial performance and regional development. The compactness, connections, spatial distances and other spatial characteristics of cities also influence the availability of spillover effects and thus determine differences in entry and exit conditions for development (Krmenc & Esparza, 1999). For example, the compact city benefits work productivity and economic performance (Cervero, 2001); dispersion between cities will promote the internalization of knowledge spillovers and lead to the emergence and development of oligopolistic or monopolistic firms (Glaeser, Kallal, Scheinkman, & Shleifer, 1992; van der Panne, 2004); the rise of urban agglomeration transportation networks such as expressways, high-speed rail systems and airlines contributes to the shortening of spatial distance and provides opportunities for the development of horizontal and polycentric city networks and regional integration (Xu, Zhang, Zhang, Wang, & Zhang, 2019). Meanwhile, research on the 'borrow-size' effect shows that small cities in urban agglomerations more susceptible to positive externalities of industrial development and to functioning facilities in large cities than other isolated small cities (Alonso, 1973), and recent studies have found that larger cities are more likely to benefit from the "borrow-size" effect (Meijers & Burger, 2016). Although these fundamental and rigorous works provide a large collection of valuable results and have expanded understanding, they are based on cross-sectional data. Due to the path dependence of regional development, it remains unclear what role the characteristics of industrial development play in urban or regional

spatial structures and intercity relations and how such features impact the spatial integration and evolution of urban agglomerations, especially given that industries have experienced tremendous changes over the past decades and are entering a new industry 4.0 era (Salgues, 2018).

2.3. Development of Chinese urban agglomerations

With national reform and opening policies applied in 1978, while Chinese cities have experienced dramatic patterns of urban expansion, economic growth and migration, uneven development and high environmental costs must also be addressed (He, Chen, Mao, & Zhou, 2016; Wei et al., 2017; Wei, 2012). Unlike building mechanisms of urban agglomeration adopted in the UK and US, China has adopted a composite mechanism. Differences in urban agglomeration development stages, development mechanisms and core issues render many theories developed in the North American and European contexts unable to directly describe the evolution of urban agglomerations and support only inferences on the coordinated development of Chinese urban agglomerations (Huang, Li, & Hay, 2016). With changes in spatial structures and forms, urban development strategies and development preferences at different stages still play a significant role. Meanwhile, the game between local cities and government intervention have rendered spatial structures not fully self-organized and have fragmented functional economic connections (Innes, Booher, & Di Vittorio, 2010; Li & Wu, 2017). In addition, the Chinese government has proposed several development plans for promoting regional integration in urban agglomerations, and some planning outlines have been put forward to achieve coordinated development on larger scales such as in the YREB and Guangdong-Hong Kong-Macao Greater Bay Area. However, while urban agglomeration has been actively embraced, debates continue on the effectiveness and rationality of planning and specific measures. Therefore, as a laboratory to examine contemporary urban changes and planetary urbanization, research on China's urbanization will not only provide valuable knowledge for the Global South, but will also contribute to Western urban research on neoliberalism in general and on urban entrepreneurialism in particular (Wu, 2015, 2017).

3. Study area and datasets

3.1. Study area

The YREB covers a broad area that includes two municipalities and nine provinces and that accounts for 21% of China's total area and for more than 40% of China's population and gross domestic product (GDP). Significant differences exist between different municipalities and provinces and can reflect the various characteristics and patterns of regional development observed in China (Fig. 1). The Outline of Yangtze River Economic Belt Development Planning was released in 2016. The plan proposes relying on shipping functions of the Yangtze River, constructing a comprehensive three-dimensional traffic corridor, guiding industry transfer from eastern to western parts of the country, and promoting coordinated development in the Yangtze River Basin. As a national development strategy adopted in China, the plan will benefit the further development of cities in the Yangtze River Basin and create new opportunities for urbanization and economic development in China.

As a core area of the YREB, the three main urban agglomerations located to the east and west will become growth engines for the YREB. Therefore, based on the Outline of Yangtze River Economic Belt Development and other related development planning documents, the authors selected three urban agglomerations and 11 metropolitan areas and 43 cities within them as study objects (Table 1).

Since the second turning point of urbanization appeared was initiated in 1995, Chinese urbanization has accelerated further. The average growth rate has remained at 1.4% per year ever since (Chen, Liu, & Tao, 2013). Therefore, we apply 1995 as the base period and the cycle (five years) of the National Economic Development Plan as a temporal interval.

3.2. Study datasets

In this study, nighttime light data is mainly used to explore the spatial integration of urban agglomerations. Industrial enterprise dataset is employed to measure the characteristics of industrial development. The details of these two datasets are explicated as follows:

Nighttime light data could obtain from DMSP/OLS and NPP/VIIRS datasets, which covers 1993–2012 and 2012 to the present, respectively. Nighttime stable light data (NSL) include lights from cities, towns, and other sites with stable lighting and exclude light emitted due to ephemeral factors such as fishing boats and fires, and they can be directly obtained from the National Centers for Environment Information (NCEI, <https://ngdc.noaa.gov/eog/download.html>) of the National Oceanic and Atmospheric Administration (NOAA). The data sources, structures and radiometric resolutions of NPP/VIIRS data are distinct from those of DMSP/OLS data. DMSP/OLS data are obtained from different satellites and OLS data are subjected to no-board calibration, creating discrepancies in digital numbers (DNs) and the number of lit pixels generated from different satellites for the same year. Abnormal fluctuations in DNs also appear as well as in the number of lit pixels collected from a same satellite for different years. However, great efforts have been made to improve upon the comparability and continuity of DMSP/OLS and NPP/VIIRS data. Preprocessing involved the following three tasks:

- (1) The unity of data structures for DMSP/OLS and NPP/VIIRS was applied, including changes made to pixel types and depths to the same data standards (Li, Li, Xu, & Wu, 2017) and the setting of the resample pixel size to 1 km.
- (2) The correction for DMSP/OLS and NPP/VIIRS data following Elvidge and Zhifeng Liu's approach (Elvidge et al., 2009; Liu, He, Zhang, Huang, & Yang, 2012) with three steps: **a. Intercalibration.** A second-order regression model was developed to improve the comparability of annual DMSP/OLS and NPP/VIIRS data by comparing DN values of the reference data (DMSP satellite F16 in 2007 with the highest cumulative DN value) and remaining images for the city of Jixi in Heilongjiang Province. **b. Intra-annual composition.** This process mainly focused on DMSP/OLS data. A lit pixel was defined as an intra-annual unstable lit pixel if it was detected by only one satellite. When a lit pixel was detected by only one satellite, it should be replaced with the background value; otherwise, the DN value of each intra-annual stable lit pixel was replaced by the averaged DN value of two NSL images from the same year. **c. Interannual series correction.** We assumed that lit pixels should become brighter overtime and that patches detected in an early NSL image should not disappear in the YREB. An interannual series correction was applied to enhance the continuity of the DMSP/OLS and NPP/VIIRS data over the multi-year period.
- (3) The extraction of urban space from NSL data. Scholars have proposed a few ways to extract information on urban land and human activities, including the threshold extraction method (Elvidge, Baugh, Kihn, Kroehl, & Davis, 1997; Imhoff, Lawrence, Stutzer, & Elvidge, 1997), machine learning classification (Cao, Chen, Imura, & Higashi, 2009) and cluster-based techniques (Zhou et al., 2014). The threshold technique using with ancillary data to extract urban space was employed in this study. Urban land included in the land

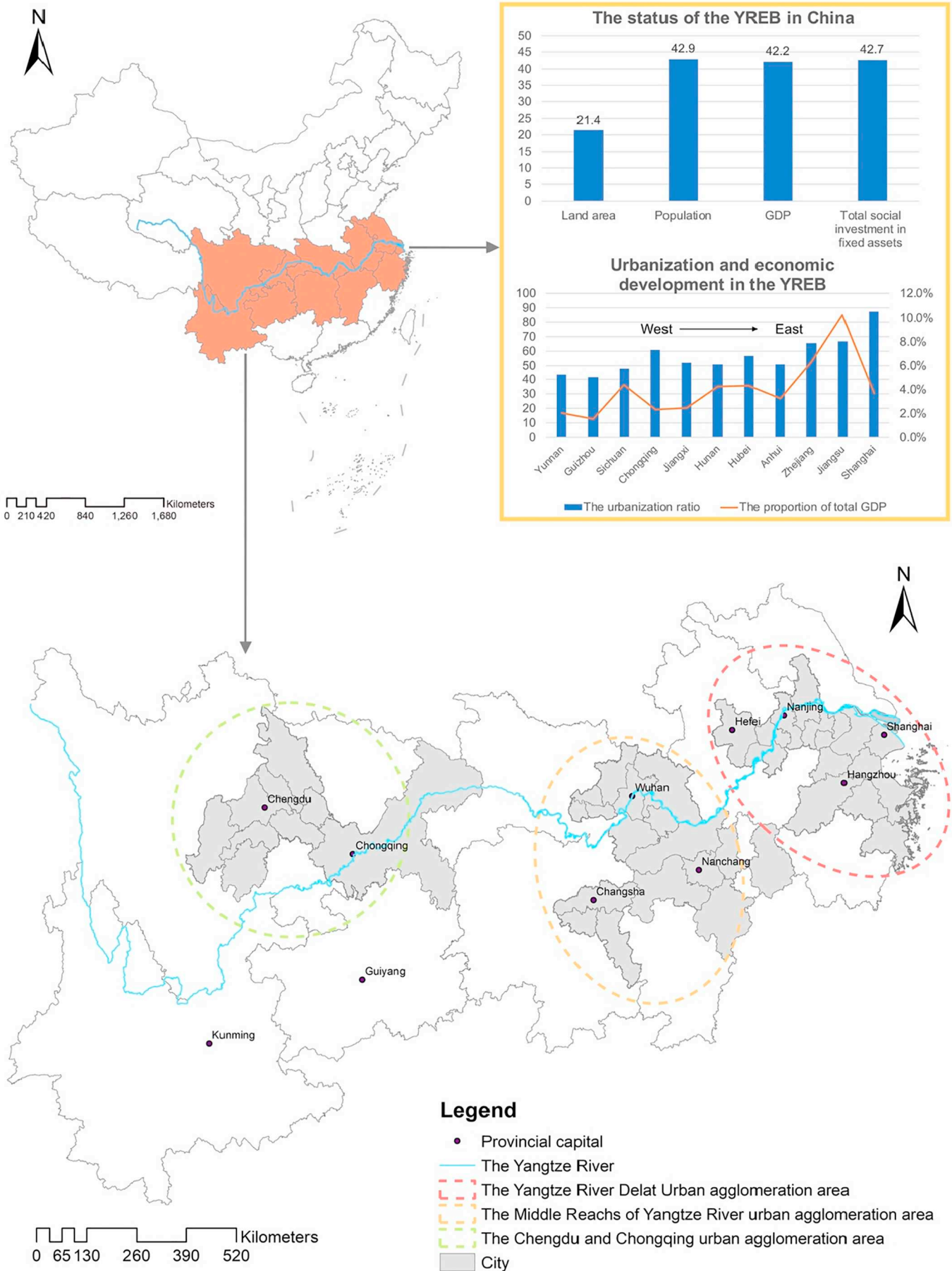


Fig. 1. The location and boundaries of the study area.

Table 1

The composition of urban agglomerations and metropolitan areas.

Urban agglomeration	Metropolitan area	City
The Yangtze River Delta urban agglomeration (YRD)	Shanghai metropolitan area Suxichang metropolitan area Nanjing metropolitan area Hangzhou metropolitan area Ningbo metropolitan area Hefei metropolitan area	Shanghai Suzhou; Wuxi; Changzhou Nanjing; Zhenjiang; Yangzhou Hangzhou; Jiaxing; Shaoxing; Huzhou Ningbo; Taizhou; Zhoushan Hefei; Wuhu; Maanshan
The middle reaches of the Yangtze River urban agglomeration (MRYS)	Wuhan metropolitan area Changzhutan metropolitan area	Wuhan; Ezhou; Huangshi; Huanggang; Xiaogan; Xianning; Tianmen; Qianjiang; Xiantao Changsha; Zhuzhou; Xiangtan
The Chengdu and Chongqing urban agglomeration (C&C)	Nanchang metropolitan area Chengdu metropolitan area Chongqing metropolitan area	Nanchang; Jiujiang; Shangrao; Fuzhou; Yichun Chengdu; Deyang; Mianyang; Meishan; Leshan; Ziyang; Suining; Yaan Chongqing

Note: The column of City includes municipalities, prefectural-level cities and county-level cities(provincial-controlled).

use dataset of the Data Centre for Resources and Environmental Sciences of the Chinese Academy of Sciences (RESDC) (<http://www.resdc.cn/>), a reliable and accurate ancillary dataset, was chosen as a reference dataset. To more reasonably extract thresholds, the YREB was divided into different regions by levels of economic development, and optimal thresholds were determined for them. An optimal threshold for each region was applied such that the lit area at the given threshold closely matched the number of urban land use pixels included in the reference dataset.

Chinese industrial enterprises datasets generated from industrial survey statistics of the National Bureau of Statistics was employed. The survey covers industry types; the gross value of industrial output; employment, production and sales trends; costs and expenses; the capability of inventory and production and other data. As the available data only cover 1995–2013, 2013 industrial enterprise data are used to proxy the data for 2015, while data for the other years are not affected. In addition, data for different years are inconsistent for certain cities due to administrative adjustments. We thus recalculate the data for these cities according to current administrative divisions. Other socio-economic data are taken from the China City Statistical Yearbook for the corresponding years.

4. Methods

4.1. Quantifying the spatial integration of urban agglomerations

4.1.1. Identifying the city origin centre

The city origin centre is a start point of urban development. To quantify urban spatial evolution, an approximate start point should be determined first. The identification of the city origin centre will directly affect the accuracy of sector analysis results. Numerous studies have used administrative (Wu, Zhao, Zhu, & Jiang, 2015), CBD (Pez, 2012) or geometric centres (Ji, Ma, Twibell, & Underhill, 2006) as the city centre, but the city centre identified by these methods cannot serve as a reliable proxy of an urban development's starting point and core areas. Therefore, multi-year NSL data are used to identify city origin centre following three identification rules: (1) NSL data for 1995, 2000, 2005, 2010, and 2015 are used to identify areas with the highest DN values in each city. (2) Areas with the highest DN values for each year are matched. When the areas with the highest DN values for each year overlap, the overlapping areas are intercepted; when they do not overlap, the area with the highest value of the earlier year is applied. (3) The geometric centre of the area with the highest value determined in previous steps is used to represent the city origin centre.

4.1.2. Sector analysis and urban form

Sector analysis involves the division of space into a number of sector areas with equal central angles based on the city origin centre to reflect urban spatial changes oriented in different directions. In this work, city origin centres are determined with the above method; based on orientation relationships for each city, urban space extracted from NSL and administrative areas are divided into 16 parts; urban development intensity (UDI_i) is calculated for each direction. The equation is as follows:

$$UDI_i = \frac{A_{urban(i)}}{A_{admin_developable(i)}} \quad (1)$$

where i is the direction, and A_{urban} denotes the urban space. $A_{admin_developable}$ is the land area suitable for urban development and is obtained by removing water bodies and land with a slope of greater than 25°.

The standard deviation (SD) of UDI_i in 16 directions is calculated to reflect the urban form and its changes. It should be noted that the broad meaning of urban form is adopted, which includes the spatial structure and formation of a city's physical environment and socio-economic activities (Batty, 2008). When the SD is smaller, the urban form is simpler and development occurring in all directions is more balanced; when the SD is larger, the urban form is more complex and development levels in each direction clearly vary. The following formula is used:

$$SD = \sqrt{\frac{\sum_{i=1}^{16} (UDI_i - \overline{UDI})^2}{15}} \quad (2)$$

4.1.3. Measuring intercity relations and their changes

Intercity relations aim to reflect the spatial tendency and preference of a city towards its geographical neighbours. We first determine the orientation relationship between neighbouring cities, sum the UDI_i according to the identified orientation relationship and calculate its ratio to the total UDI for a given city. CityRelation can reflect the strength of development preferences observed between a certain city and a neighbouring city, which is directional and differs in either direction. The following formula is used:

$$CityRelation = \frac{\sum_{i=1}^n UDI_i}{\sum UDI_i} \quad (3)$$

The growth rate (AGR) of UDI for a certain direction will be higher or lower than the overall AGR. When the AGR of a certain direction is higher (lower) than the overall AGR, the direction presents some (dis)

advantages and attractions (exclusions) for urban development. Therefore, the AGR of UDI for a certain direction is calculated and compared to the overall AGR.

$$AGR = 100\% \times \left[\left(\frac{UDI_{end}}{UDI_{start}} \right)^{1/d} - 1 \right] \quad (4)$$

where the AGR (%) is the growth rate of UDI. UDI_{start} and UDI_{end} represent the UDI measured in the starting and ending periods, respectively, and d is the time interval employed.

When the AGR of a certain direction is higher than a city's overall AGR, the city is identified as approaching its neighbouring city. When the AGR of a certain direction is lower, the city is designated as lagging. When the AGR of a certain direction is 0, a city is defined as stagnant relative to its neighbouring city. The spatial evolution of intercity relations manifests in six ways: approach-approach, approach-lag, approach-stagnation, lag-lag, lag-stagnation, and stagnation-stagnation.

4.2. Measuring the characteristics of industrial development

To measure Marshallian specialization, Jacobian diversification, local competition and industrial structure similarities, four indexes are employed to measure the characteristics of industrial development. 581 4-digit industries in the Chinese Industrial Enterprise Database are used to calculate these four indexes for 1995 to 2015. The 4-digit industries are the smallest units in the National Economic Industry Classification.

4.2.1. The urban production structure specialization index

To measure Marshallian specialization in a city, the urban production structure specialization index (PS_j), the most widely used index, is calculated by averaging the specialization of various industries within a city. The equation adopted is as follows:

$$PS_j = \frac{\sum_i PS_{ij}}{n} \quad (5)$$

$$PS_{ij} = \frac{\frac{E_{ij}}{\sum_i E_{ij}}}{\frac{\sum_i E_{ij}}{\sum_i \sum_j E_{ij}}} \quad (6)$$

where i is a 4-digit industry, j is a city in the YREB, n is the number of 4-digit industries, and E is employment.

4.2.2. The urban production structure diversification index

The urban production structure diversification index does not traditionally consider whether there is a correlation between industries, directly uses the proportion of each industry as weight, and integrates the entropy values of all industries (Frenken et al., 2007). Urban production structure diversification calculated with this method can be divided into related and unrelated varieties. In this work, related variety is measured as the weighted sum of entropy at the 4-digit level for each 2-digit class (Ercole & O'Neill, 2016) and is measured as follows:

$$PD = \sum_{g=1}^G P_g \left(\sum_{i \in S_g} \frac{P_i}{P_g} \log_2 \left(\frac{1}{P_i/P_g} \right) \right) \quad (7)$$

$$P_g = \sum_{i \in S_g} P_i \quad (8)$$

where S_g is a 2-digit sector, P_i denotes the share value of 4-digit industrial employment, and P_g is the sum of 4-digit shares P_i .

4.2.3. The degree of local competition

The degree of local competition ($Comp_{ij}$) measures the number of firms per worker per industry i per region j relative to the overall equivalent. Related low values can be explained with Marshall's theory as denoting larger average firm sizes and levels of market power;

related high values are associated with increased levels of industry-specific local labour market competition following Jacobs' conclusions (van der Panne, 2004). $Comp_j$ can be measured with the following equation:

$$Comp_j = \frac{\sum_i Comp_{ij}}{n} \quad (9)$$

$$Comp_{ij} = \frac{\frac{Firms_{ij}}{E_{ij}}}{\frac{\sum_i \sum_j Firms_{ij}}{\sum_i \sum_j E_{ij}}} \quad (10)$$

where $Firms$ denotes the total number of firms.

4.2.4. The degree of industrial structure similarity

The degree of industrial structure similarity (ISS) is used to quantify the common characteristics of industrial structures observed across different cities. In the present work, the ISS proposed by Krugman is employed (Krugman, 1993). As the Krugman index (K) has a value range of $[0, 2]$, when industrial structures are more (dis)similar between cities, its value is closer to 0(2). To facilitate the interpretation for our regression results, a reciprocal is used. Industrial structure similarity is measured as follows:

$$ISS = \frac{1}{K} = \frac{1}{\sum_{i=1}^n |S_{1i} - S_{2i}|} \quad (11)$$

where S_{1i} and S_{2i} represent the gross value of the industrial output of cities 1 and 2, respectively.

4.3. Spatial correlation and panel regression analysis

4.3.1. Bivariate Moran's I

Bivariate Moran's I is used to explore spatial clusters and spatial dispersion between specialization/diversification and urban forms, and global and local Bivariate Moran's I values are used for this purpose. The global bivariate Moran's I is used here to examine whether and to what extent spatial correlations between specialization/diversification and urban form exist across the YREB; the local bivariate Moran's I is used to draw spatial correlations within different cities. The equation is used as follows:

$$I = \frac{n \sum_{i=1}^n \sum_{j=1}^n W_{ij} \left(\frac{x_{i,c} - \bar{x}_c}{\sigma_c} \right) \left(\frac{y_{j,s} - \bar{y}_s}{\sigma_s} \right)}{(n-1) \sum_{i=1}^n \sum_{j \neq i}^n W_{ij}} \quad (12)$$

$$I' = \frac{x_{i,c} - \bar{x}_c}{\sigma_c} \sum_j W_{ij} \left(\frac{y_{j,s} - \bar{y}_s}{\sigma_s} \right) \quad (13)$$

where I and I' represent the global and local bivariate Moran's I for specialization/diversification and urban form, respectively. n denotes the total number of cities in the YREB. W_{ij} is a spatial weight matrix for measuring spatial correlations between cities i and j based on geographic distance with a threshold of 600 km, which is a reasonable distance that could cover the distances between any two cities within an urban agglomeration. $x_{i,c}$ and $y_{j,s}$ represent the specialization/diversification and urban form of city i , respectively. σ_c and σ_s denote the standard deviation of specialization/diversification and urban form, respectively. The range of I and I' is $[-1, 1]$ where a positive value indicates that a higher PS/PD is more easily surrounded with cities with more complex urban forms. Inversely, a negative value indicates that a higher PS/PD is more easily surrounded by cities of more simple urban forms. When the absolute values of I and I' are closer to 1, they have a stronger spatial correlation. In addition, a cluster map is generated through the local bivariate Moran's I calculation to reflect the spatial correlation between specialization/diversification for a specific city and the urban form of neighbouring cities.

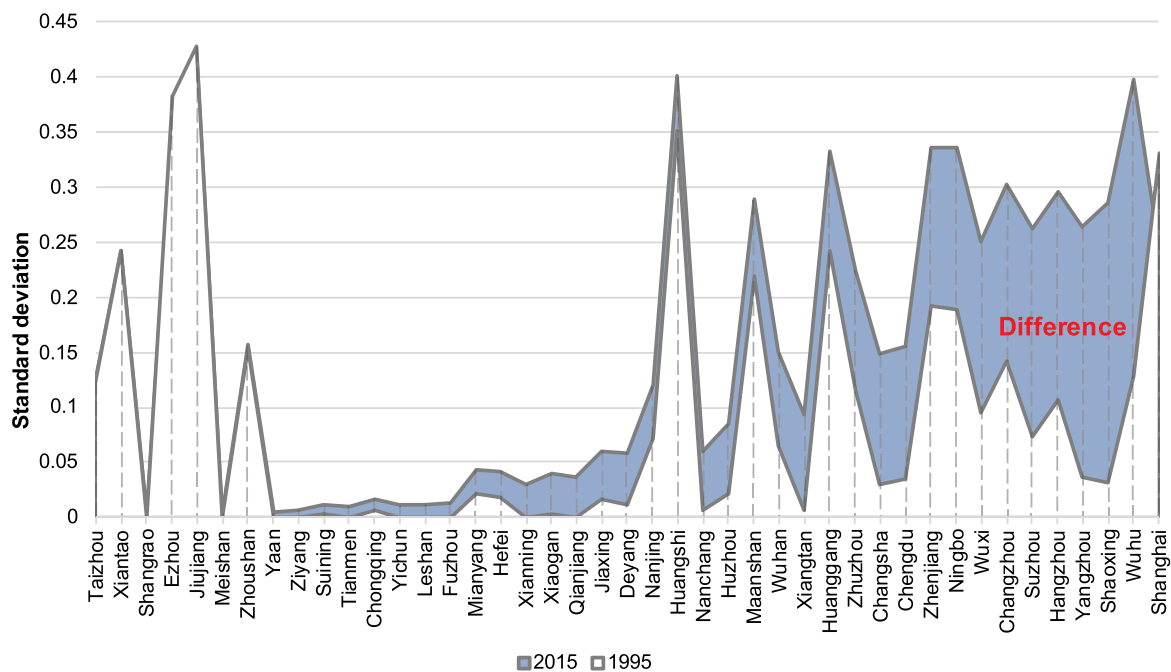


Fig. 2. The SD of UDI in 16 directions for the YREB.

4.3.2. Panel regression

A panel regression model is employed to analyse the relationship between intercity relations and the characteristics of industrial development. The sample consists of a pair of spatial adjacent cities. In this study, the panel regression includes 71 cross-sections (N) and 5 time series (T). It is a micropanel, and the time intervals are equal. For micropanels where T is relatively small, non-stationarity is not considered in micropanels (Baltagi, 2008). In addition, the unit root tests developed under the assumption of a large T are usually unsatisfactory because of size distortions (Hadri & Larsson, 2005). Therefore, unit root tests will not be performed in this study. According to the Hausman test, our regression applied the individual fixed effect model. To ensure the robustness of our regression results, heteroskedasticity correction was performed. Specifically, the following regression model was employed:

$$\begin{aligned} \ln(\text{CityRelation})_{it} = & \alpha_i + \beta_0 + \beta_1 * \ln(\text{PS})_{it} + \beta_2 * \ln(\text{PD})_{it} + \beta_3 \\ & * \ln(\text{Comp})_{it} + \beta_4 * \ln(\text{ISS})_{it} + \beta_5 \\ & * \ln(\text{PS} * \text{Comp})_{it} + \beta_6 * \ln(\text{PD} * \text{Comp})_{it} + u_{it} \end{aligned} \quad (14)$$

It should be noted that our means of calculating dependent and independent variates is not traditional, as our sample includes pairs of spatially adjacent cities where β_i denotes the constant term ($i = 0$) and the coefficients to be estimated. PS, PD, and Comp are measured by averaging the index for a pair of cities.

5. Results

5.1. The spatial integration of urban agglomerations

5.1.1. Changes in urban form

From 1995 to 2015, changes in urban spatial form show clear differences in the YREB (Fig. 2). Overall, the urban form is more complex than before. The magnitude of change observed in the eastern urban agglomeration is greater than that observed in the central and western cities, and changes observed in provincial capital and primary cities are

more intense than those of ordinary cities. It should be noted that as a special case, differences in development intensity observed in all directions in Shanghai are less significant than those of the former. As a product of urban and regional development, urban form is influenced by economic growth, industrial transformation and the role of urban and social systems and responds to these transformations through morphological changes.

City ranks and GDP increments are used to classify SD differences for further analysis (Fig. 3). In general, the economic growth of cities showing strong changes in SD is more pronounced than that of cities showing less significant changes. Compared to those of the other regions, the economic increments of cities are higher in the YRD, and changes in SD are more pronounced than average levels observed for the YREB. Cities in the multicentric metropolitan area also exhibit these characteristics. Except in Nanjing, Hefei, Nanchang and Chongqing, the change magnitude of SD above average in other primary cities. In initial stages of urban development and in the pre-industrialization period, industrial development largely maintains the local economy. Industrial characteristics render development requirements less rigorous, and the proportion of external economic activities is generally low. However, with rapid industrialization, the appropriate geographical location will facilitate the acquisition of production materials and spur spatial agglomeration so that firms of the same sector can promote economies of scale, knowledge spillovers and other positive externalities. In turn, functionality and economic efficiency are more advantageous in specific directions or regions. The city will thus pursue rapid development in this direction to maximize the utility and efficiency of input resources, which will also transform urban forms from simple to more complex and lead to further alienation. In other words, economic growth as the process and production of local industrial arrangement, development and evolution will have a certain correlation with changes in urban form. In addition, exceptional socio-economic foundations, favourable discourse and the legislative power of local governments affect a city's status within its network and the resources obtained. This is why urban forms change more and grow more complex in the provincial capital, primary and eastern cities studied.

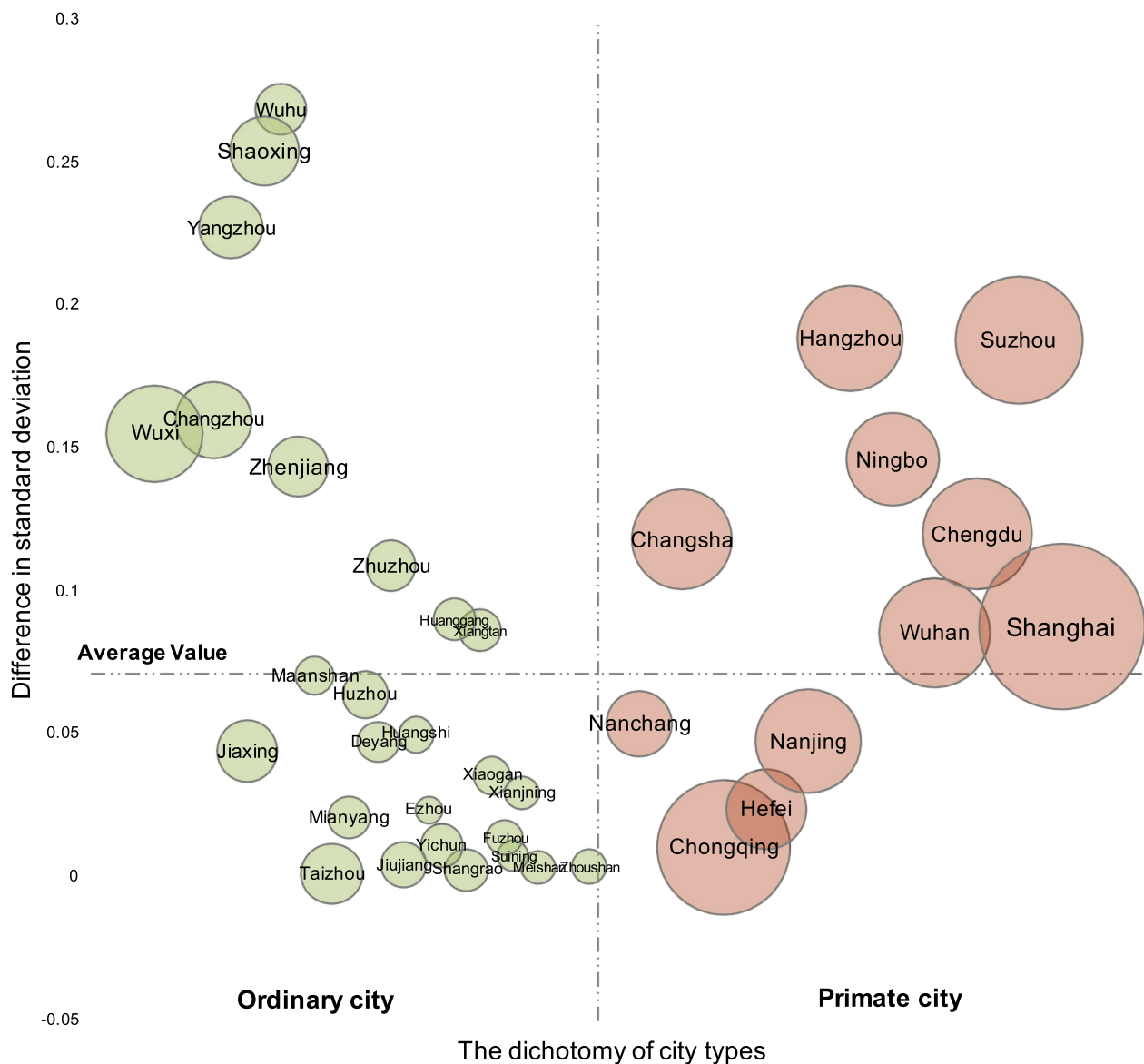


Fig. 3. A typology of urban form changes.

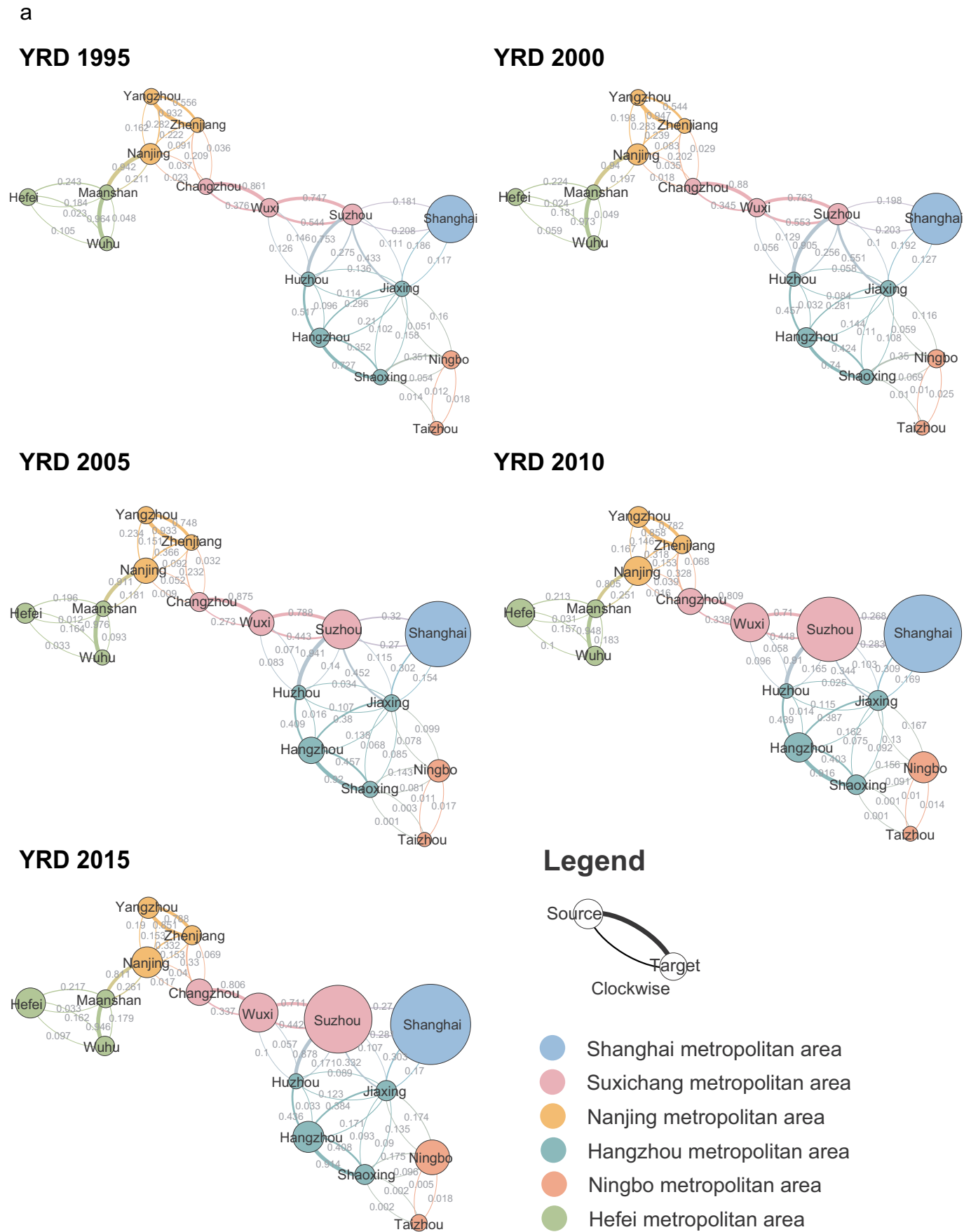
Notes: (1) Individual bubbles represent cities. (2) The vertical axis denotes the difference in SD values observed between 1995 and 2015; the horizontal axis denotes the dichotomy of city types with colours used to highlight this dichotomy. (3) Bubble sizes are proportional to GDP increments for 1995 to 2015.

5.1.2. Measuring intercity relations

Intercity relations show clear differences in intensity as well as regional and model differences for the YREB. In three urban agglomerations, relations between metropolitan areas and cities in the YRD are stronger than those of the MRYR and C&C shown in Fig. 4 (a–c). Among them, the YRD presents superior natural conditions, a sound foundation for socio-economic development, and cities are positioned relatively close together, which is conducive to spatial agglomeration and spillover effects. Thus, the relations between various metropolitan areas are relatively strong. The links between primary and ordinary cities are more balanced than those of other urban agglomerations, and the differences observed in both directions gradually decline. As the development of the YRD appears as a “bottom-up” model, local industries show strong patterns of development momentum and high levels of agglomeration, strengthening relations between ordinary cities in various metropolitan areas. In addition, some ordinary cities prefer to link with the primary city in the neighbouring metropolitan area rather than the primary city in the same metropolitan area (e.g., Maanshan and Huzhou). This phenomenon shows that the influence of certain primary

cities crosses provincial administrative boundaries and promotes spatial integration and strengthened coordinated development between different provinces.

However, the characteristics and transformation of intercity relations in the MRYR and C&C present different features. In these urban agglomerations, intercity relations are relatively weak and especially between different metropolitan areas. With the exception of the Changzhutan metropolitan area, the other metropolitan areas follow a typical mononuclear model. In turn, primary cities show more balanced development in all directions, and the relations of ordinary cities in the same metropolitan area continue to build and strengthen. Given their dominant status in the urban network, the speed and scale of the development of primary cities are greater than those of ordinary cities. Moreover, intercity relations are imbalanced between some primary cities and ordinary cities, as agglomeration effects are stronger than diffusion effects. To avoid siphoning effects, some ordinary cities do not approach the primary city. Furthermore, due to their weaker development foundations and less favourable external conditions, intercity relations are not stable and associated trends fluctuate.



(caption on next page)

Fig. 4. (a) Intercity relations in the YRD. (b–c). Intercity relations in the MRYR and C&C.

Notes: (1) Individual circles represent cities. (2) The line width between the two circles denotes the strength of intercity relation, and highlighted by corresponding CityRelation values. (3) The size of the circle is proportional to the sum of DN values in a city for the corresponding years.

5.1.3. The spatial evolution of intercity relations and regional spatial integration

The spatial evolution of intercity relations involves regional spatial integration and may involve other influencing factors. Fig. 5 (a–c) illustrates the spatial evolution of intercity relations in three urban agglomerations for four consecutive periods classified by city rank and metropolitan area. In general, intercity relations are oriented towards the approach and promote regional spatial integration. The spatial evolution of intercity relations can be described as follows. (1) Overall, in initial stages of urban development, while rates of change in a certain direction are high or low in the YRD, stagnation is less common in this region. However, the opposite pattern is observed in the MRYR and C&C. (2) Approach-approach patterns mainly appear between primary and ordinary cities within the same metropolitan area of the YRD in earlier periods. With the strengthening of intercity relations and regional spatial integration, the links between different metropolitan areas have been improved. Especially since 2005, the relation between primary and ordinary cities in different metropolitan areas has grown stronger. Primary cities are radiating towards ordinary cities not located in the same metropolitan area or even the same province. (3) In the MRYR, the Wuhan and Nanchang metropolitan areas exhibit a typical mono-nuclear model. As the primary cities in each metropolitan area are larger in scale, are ranked higher and present more dispersed spatial distributions, approach-approach patterns mainly appear between primary and ordinary cities of the same metropolitan area. Primary cities dominate these relations. Due to the single industrial structures and lagging development of ordinary cities, spatial integration mainly depends on the development of primary cities of metropolitan areas in the MRYR. (4) Prior to 2005, intercity relations between cities in the C&C were largely lagging or stagnating. After this point, intercity relations gradually strengthened and spatial integration emerged in the northern area of the C&C rather than across the entire urban agglomeration.

Furthermore, the spatial evolution of intercity relations is fluctuating or lagging (e.g., the opposite intercity relations between two cities are observed in a consecutive period such as in (a, l) and (l, a)). On the one hand, while the two cities seek to achieve stronger relations and approach one another, they are not fully synchronized with the development process due to the influence of industrial arrangements, infrastructure construction and local administrative decision-making. On the other hand, competition between cities may also slow or stagnate spatial integration, especially in the context of China. The hierarchical nature of China's administrative divisions and the effects of fiscal decentralization have created relatively weak horizontal relations. In addition, the relations between provincial governments lack an effective coordination mechanism, which cannot directly intervene or regulate processes of competition and integration and which allow this uncoordinated phenomenon to develop further. Under the effects of such institutional settings and scale-building mechanisms, intercity relations will clearly change with the transformation in city interests.

5.2. The relationship between spatial integration and the characteristics of industrial development

5.2.1. Spatial correlations between industrial specialization/diversification and urban forms

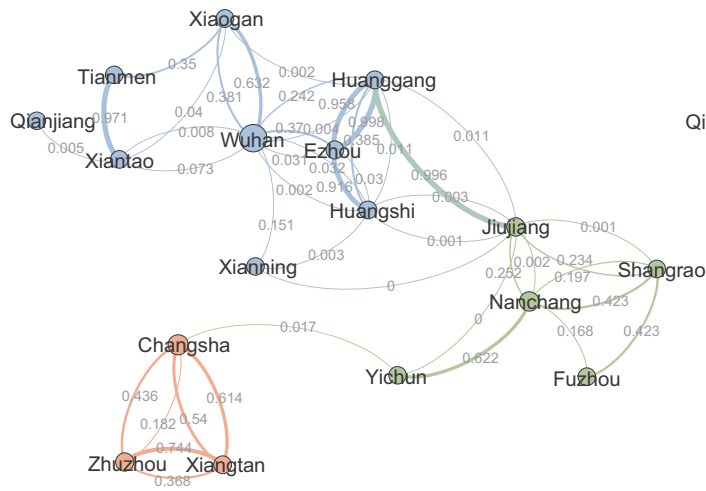
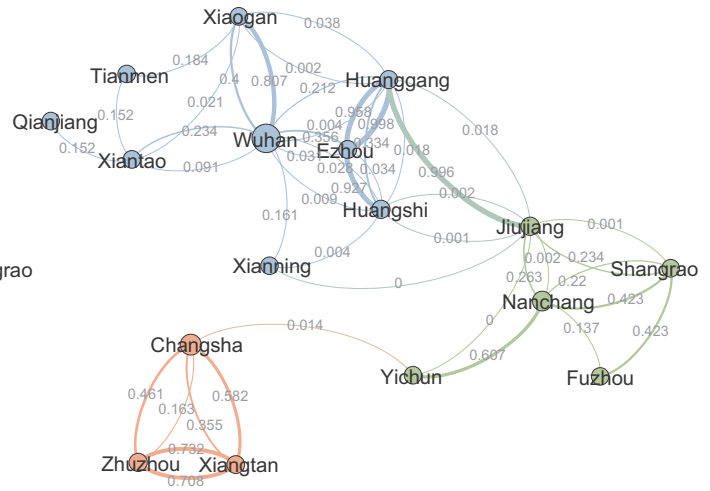
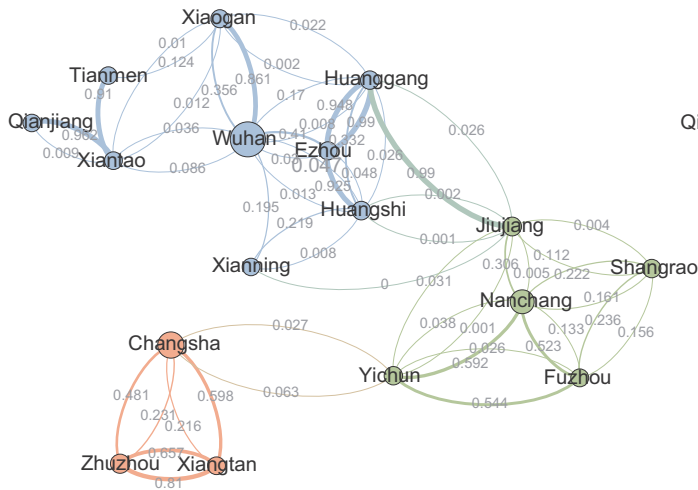
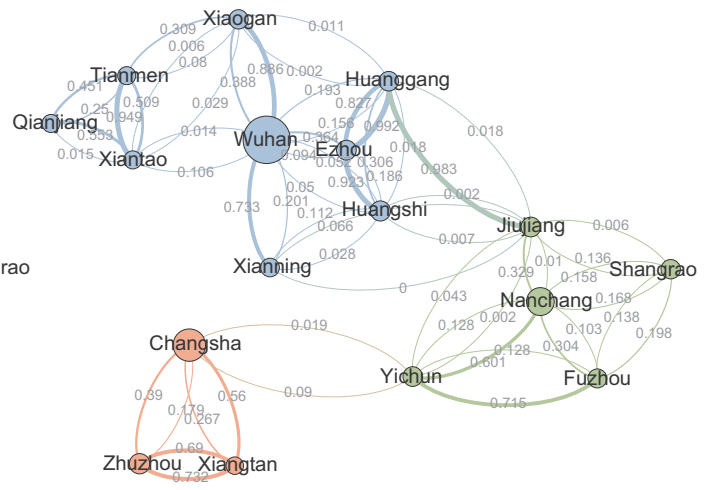
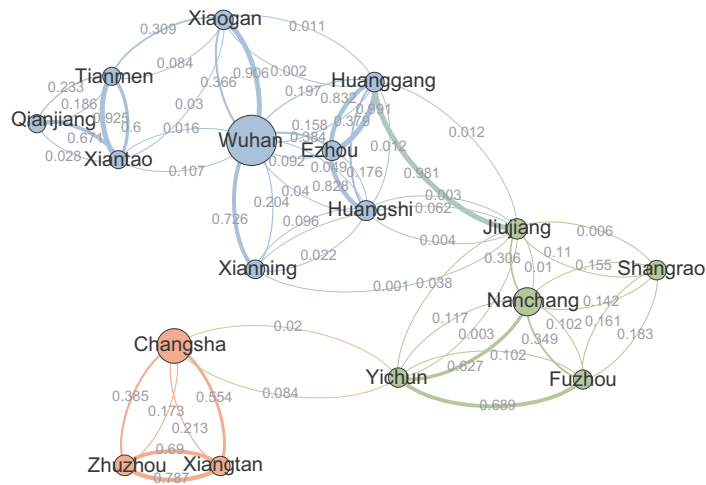
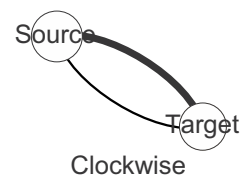
The spatial correlation between industrial characteristics and urban forms is analysed from the bivariate Moran's I. As industrial competition and urban form do not show significant correlations for the five studied time periods, this section mainly focuses on spatial correlations between industrial specialization/diversification and urban forms.

According to the results given in Table 2, *P*-values of all the models are less than 0.05, and there is a significantly negative (positive) correlation between industrial specialization (diversification) and urban form for 1995 to 2015. The bivariate LISA map shown in Fig. 6 (a–b) reveals four types of spatial correlations between industrial specialization/diversification and urban form.

Different spatial clustering patterns of industrial specialization and urban form were observed in the three urban agglomerations. (1) LH dominated in the YRD, and it did not undergo any apparent changes from 1995 to 2015. As the earliest region of China opening up to the outside world, the YRD has experienced many industrial transformations and its industrial structure has remained relatively balanced throughout its development. The industrial sector accounts for a relatively small proportion of total employment levels, keeping levels of industrial specialization low. “Bottom-up” urbanization has strengthened linkages between local cities. In addition, infrastructure is developing rapidly to form a complete and competitive transportation network, which has benefitted regional spatial integration and coordinated development. This has led to variations in the UDI in different directions, and urban forms have grown more complex. (2) Spatial clustering patterns observed in the MRYR are dominated by HH; HH is continuously strengthened in this region and mainly in the eastern part of the Wuhan and Nanchang metropolitan areas. Primary cities in the MRYR present LH or insignificant patterns, which is likely related to development models of the three metropolitan areas. Wuhan and Nanchang, as primary and provincial capital cities, are better able to access various forms than ordinary cities in the same metropolitan area. To support socio-economic development, such cities must exert their influence to agglomerate various production factors and direct policy-making, causing the SD of UDI to be lower. Moreover, in the initial stages of development, the Changzhutan metropolitan area presents a multi-centre structure. However, Changsha has developed in a manner similar to Wuhan and Nanchang by virtue of the advantages of primary and provincial capital cities enjoyed in recent years. (3) The C&C shows completely different spatial patterns mainly based on HL and LL with obvious variations observed within the urban agglomeration. In the Chongqing metropolitan area and the northern section of the Chengdu metropolitan area, economic development has been more rapid and industrial development are more diversification than those observed in the other areas, rendering industrial specialization less prominent. In addition, cities are positioned farther apart, and the development of infrastructure has lagged behind that of the eastern region. Clusters and spatial spillover effects of industrial development are difficult to achieve, and external relations between cities remain weak, rendering urban forms in the area simpler.

Several additional findings were also observed from the spatial correlations between industrial diversification and urban forms. (1) Almost all cities in the YRD exhibit HH, and namely, cities showing higher levels of industrial diversification are surrounded by cities characterized by complex urban forms. Local connections and industrial divisions between cities such as Yangzhou and Zhenjiang have rendered the development of cities in specific directions more obvious, while spatial integration has grown more pronounced. Areas of southern Jiangsu and northern Zhejiang have also been influenced by Shanghai, and the development of traditional light and high-tech industries has promoted industrial diversification in these regions. Therefore, HH is largely concentrated in Shanghai and in the Jiangsu and Zhejiang Provinces. Although Hefei also shows signs of HH after 2005 due to its status as a primary city provincial capital city, Wuhu and Maanshan in the Hefei metropolitan area largely exhibited LH until 2015. Major

b

MRYR 1995**MRYR 2000****MRYR 2005****MRYR 2010****MRYR 2015****Legend**

- Wuhan metropolitan area
- Changsha metropolitan area
- Nanchang metropolitan area

Fig. 4. (continued)

C

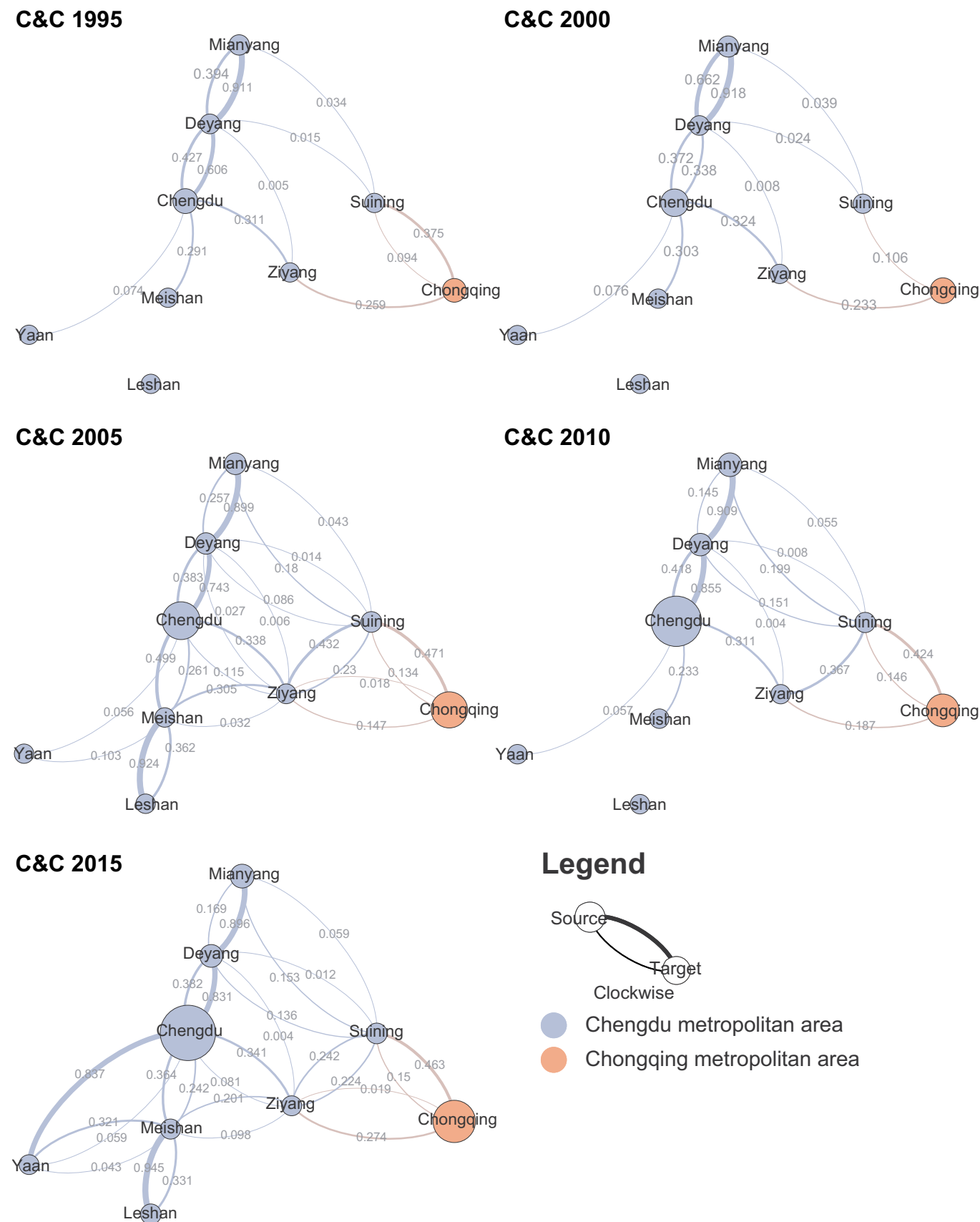


Fig. 4. (continued)

YRD (a)

City Relations	City-Pair	1995-2000	2000-2005	2005-2010	2010-2015
D(Primate, Primate)	(Shanghai, Suzhou)	(l, a)	(a, a)	(a, l)	(l, a)
	S(Primate, Ordinary)	(a, a)	(l, a)	(a, l)	(l, a)
	(Nanjing, Yangzhou)	(a, a)	(a, l)	(l, l)	(a, a)
	(Nanjing, Zhenjiang)	(a, l)	(a, a)	(l, a)	(a, l)
	(Ningbo, Taizhou)	(a, l)	(l, a)	(l, l)	(a, l)
	(Hangzhou, Shaoxing)	(a, a)	(a, a)	(l, l)	(a, l)
	(Hangzhou, Huzhou)	(l, l)	(l, l)	(a, l)	(l, a)
	(Hangzhou, Jiaxing)	(l, l)	(a, l)	(a, a)	(l, a)
	(Hefei, Maanshan)	(l, a)	(l, l)	(a, a)	(a, a)
	(Hefei, Wuhu)	(l, l)	(l, l)	(l, a)	(a, l)
	S(Ordinary, Ordinary)	(l, a)	(l, l)	(a, l)	(l, l)
	(Zhenjiang, Yangzhou)	(a, l)	(l, a)	(l, a)	(l, a)
	(Maanshan, Wuhu)	(a, a)	(a, a)	(a, l)	(l, l)
	(Shaoxing, Jiaxing)	(a, l)	(l, l)	(a, a)	(a, l)
	(Huzhou, Jiaxing)	(l, l)	(l, a)	(l, a)	(a, a)
	D(Primate, Ordinary)	(l, a)	(l, a)	(a, l)	(a, l)
	(Suzhou, Jiaxing)	(l, a)	(a, l)	(l, l)	(a, l)
	(Shanghai, Jiaxing)	(a, a)	(a, a)	(a, a)	(a, l)
	(Nanjing, Changzhou)	(l, l)	(a, l)	(l, a)	(a, a)
	(Nanjing, Maanshan)	(l, l)	(l, l)	(a, l)	(a, a)
	(Ningbo, Shaoxing)	(a, l)	(a, l)	(a, a)	(a, a)
	(Ningbo, Jiaxing)	(a, l)	(a, l)	(a, a)	(a, a)
D(Ordinary, Ordinary)	(Shaoxing, Taizhou)	(l, l)	(l, a)	(l, l)	(a, a)
	(Wuxi, Huzhou)	(l, l)	(l, a)	(l, a)	(l, a)
	(Changzhou, Zhenjiang)	(l, l)	(a, a)	(a, a)	(a, a)

MRYP (b)

City Relations	City-Pair	1995-2000	2000-2005	2005-2010	2010-2015
S(Primate, Ordinary)	(Wuhan, Huanggang)	(l, a)	(l, a)	(a, a)	(a, a)
	(Wuhan, Xianning)	(a, s)	(a, s)	(a, a)	(a, l)
	(Wuhan, Ezhou)	(l, l)	(a, l)	(l, a)	(a, l)
	(Wuhan, Xiaogan)	(a, a)	(l, a)	(a, a)	(l, a)
	(Wuhan, Huangshi)	(l, a)	(a, a)	(a, a)	(l, l)
	(Wuhan, Xiantao)	(a, a)	(l, l)	(a, l)	(a, a)
	(Changsha, Xiangtan)	(l, l)	(a, l)	(l, a)	(l, l)
	(Changsha, Zhuzhou)	(l, a)	(a, a)	(l, l)	(l, l)
	(Nanchang, Jiujiang)	(a, a)	(a, a)	(a, a)	(l, a)
	(Nanchang, Yichun)	(l, s)	(l, s)	(a, a)	(a, l)
D(Primate, Ordinary)	(Nanchang, Fuzhou)	(l, s)	(l, a)	(l, l)	(l, a)
	(Nanchang, Shangrao)	(a, s)	(a, l)	(l, a)	(l, l)
	(Changsha, Yichun)	(l, s)	(a, s)	(l, l)	(a, a)
	S(Ordinary, Ordinary)	(l, a)	(l, a)	(l, l)	(a, l)
	(Huangshi, Huanggang)	(a, a)	(l, a)	(l, a)	(l, l)
	(Huangshi, Ezhou)	(a, a)	(l, a)	(l, a)	(l, l)
	(Huangshi, Xianning)	(a, s)	(a, a)	(a, l)	(l, l)
	(Huanggang, Ezhou)	(l, a)	(l, l)	(a, l)	(l, a)
	(Huanggang, Xiaogan)	(l, l)	(a, l)	(l, l)	(a, l)
	(Tianmen, Xiaogan)	(s, l)	(a, l)	(l, l)	(a, a)
	(Tianmen, Xiantao)	(s, l)	(a, a)	(l, a)	(a, l)
	(Qianjiang, Xiantao)	(s, a)	(a, l)	(l, a)	(a, a)
	(Qianjiang, Tianmen)	(s, s)	(s, s)	(a, a)	(l, l)
	(Xiantao, Xiaogan)	(a, l)	(l, l)	(l, a)	(l, a)
	(Zhuzhou, Xiangtan)	(l, a)	(l, a)	(a, l)	(l, a)
	(Jiujiang, Yichun)	(a, s)	(a, s)	(a, l)	(a, l)
	(Jiujiang, Shangrao)	(a, s)	(a, l)	(a, a)	(l, l)
	(Fuzhou, Shangrao)	(s, s)	(a, l)	(l, a)	(a, l)
	(Fuzhou, Yichun)	(s, s)	(a, s)	(a, a)	(l, l)
	D(Ordinary, Ordinary)	(l, a)	(l, a)	(l, a)	(a, l)
	(Xianning, Jiujiang)	(s, a)	(s, a)	(l, a)	(l, a)
	(Huanggang, Jiujiang)	(a, a)	(a, l)	(l, l)	(l, l)

(caption on next page)

Fig. 5. (a–c) The spatial evolution of intercity relations in the YREB.

Notes: (1) Depending on whether two cities are located in the same metropolitan area, pairs of cities are divided where **S** refers to the same metropolitan area and **D** refers to different metropolitan areas. (2) **a**, **l** and **s** denote approach, lag and stagnation patterns, respectively.

differences are observed between the provinces. (2) In the MRYR, the spatial correlation transformed from the insignificant at the beginning to the LH. In the region, industrial diversification levels within the Nanchang metropolitan area were relatively low, but urban spatial forms were more complex. HH began to emerge in the Nanchang metropolitan area after 2005, which may indicate that that complex urban forms facilitate industrial diversification. Furthermore, metropolitan areas in the MRYR largely follow a mononuclear model, which is likely why the Wuhan and Changzhutan metropolitan areas are not as significant. (3) Chongqing, Chengdu and Deyang show stable signs of HL throughout the study period, while in other areas, mostly LL patterns are observed. This is likely related to natural features of the C&C urban agglomeration, which is characterized by mountainous terrain and large spatial distances between cities. Under these disadvantageous conditions, the coordinated development of industries and connections between cities have been limited, resulting in low SD values. In addition, due to their status, Chengdu and Chongqing show inherent advantages based on a collection of production factors. Numerous industries have concentrated in Chengdu and Chongqing, increasing levels of industrial diversification.

5.2.2. The relationship between intercity relations and the characteristics of industrial development

For 1995 to 2015, a panel regression is applied to the 71 pairs of cities located spatially adjacent in the YREB to analyse the relationship between intercity relations and industrial characteristics. The regression results are shown in Table 3. Specifically, model (1) estimates the impacts of industrial specialization, diversification and competition on intercity relations. Model (2) incorporates industrial structure similarities to further analyse their impacts on intercity relations. Models (3)–(5) extend model (2) by including interactions between industrial specialization/diversification and competition. In general, all the models present relatively high adjusted R-square values, and F-statistic values for all the models are significant at the 0.1% level.

The results of model (1) confirm that intercity relations and the characteristics of industrial development are negatively correlated, and

industrial diversification and competition are significant at the 10% level. The coefficient of the independent variable is used to show how the dependent variable changes with the independent variable. For example, industrial diversification has the strongest impact on intercity relations, and when industrial diversification increases by 1%, the value of intercity relations decreases by 0.3594%. It is generally believed that industrial diversification can facilitate the sharing of labour between different industries, knowledge spillovers and the rapid transfer of information. However, things are becoming complex in modern urban and regional development. Intercity relation includes connections with cities within the region (internal relations) as well as connections with cities outside the region (external relations) (Burger & Meijers, 2012). In this study, we mainly focus on the internal relation of connections between geographical adjacent cities. With the increased flow of labour, goods, capital and information and the development and improvement of the social interaction network they rely on, the relative importance of internal relations and external relations may be changing. Intercity relations are manifested not only in between geographical adjacent cities but also in visible morphological space. According to Jacobs, no city ever became successful by just interacting with its hinterland or neighbouring cities, but cities need each other and relate as networks (Jacobs, 1969). This means that the relations between some cities and their spatially non-adjacent cities are continuously strengthened through functional links and are stronger than internal relations. As cities with higher industrial diversification gradually become the core node of city networks, their external economies will arise and spread throughout the city network. In addition, regional internal resources and capital cannot constantly support the improvement of urban diversity; these cities have to seek all kinds of development resources across a wider city network. Therefore, the relationship between a city with a high level of industrial diversification and its geographical neighbours may be weakened as a result. These results support the formation of the urban network under the flow space theory and the inapplicability of the central place theory in contemporary urban and regional development (Ducruet & Beauguette, 2014; Zhang, Derudder, Wang, & Shen, 2018). On the other hand, cities with higher

C&C (c)

City Relations	City-Pair	1995-2000	2000-2005	2005-2010	2010-2015
S(Primate, Ordinary)	(Chengdu, Deyang)	(l, l)	(a, a)	(a, a)	(l, l)
	(Chengdu, Yaan)	(a, s)	(l, s)	(a, s)	(a, a)
	(Chengdu, Meishan)	(a, l)	(l, l)	(l, a)	(a, l)
	(Chengdu, Ziyang)	(a, s)	(a, s)	(l, l)	(a, l)
D(Primate, Ordinary)	(Suining, Chongqing)	(a, s)	(a, a)	(a, l)	(a, a)
	(Ziyang, Chongqing)	(l, s)	(l, s)	(a, l)	(a, l)
S(Ordinary, Ordinary)	(Deyang, Mianyang)	(a, a)	(l, l)	(l, a)	(a, l)
	(Deyang, Suining)	(a, s)	(l, s)	(l, a)	(a, l)
	(Deyang, Ziyang)	(a, s)	(l, s)	(l, l)	(a, l)
	(Leshan, Meishan)	(s, l)	(s, l)	(l, l)	(l, l)
	(Leshan, Yaan)	(s, s)	(s, s)	(a, s)	(l, a)
	(Meishan, Yaan)	(l, s)	(l, s)	(a, s)	(l, s)
	(Meishan, Ziyang)	(l, s)	(l, s)	(l, l)	(l, a)
	(Mianyang, Suining)	(a, s)	(a, s)	(a, a)	(a, l)
	(Suining, Ziyang)	(s, s)	(a, s)	(a, a)	(l, l)

Legend

Approach-Approach	Lag-Lag
Approach-Lag/Lag-Approach	Lag-Stagnation/Stagnation-Lag
Approach-Stagnation/Stagnation-Approach	Stagnation-Stagnation

Fig. 5. (continued)

Table 2

Bivariate Moran's I between industrial specialization/diversification and urban form.

	PS			PD		
	Moran's I	Z-value	P-value	Moran's I	Z-value	P-value
1995	−0.1454	−5.6392	0.001	0.0959	3.7486	0.002
2000	−0.1732	−6.7525	0.001	0.1388	5.0895	0.001
2005	−0.1689	−6.3736	0.001	0.1249	4.8156	0.001
2010	−0.2030	−7.5898	0.001	0.0496	2.1776	0.023
2015	−0.1881	−6.8764	0.001	0.1199	4.3357	0.001

levels of industrial diversification exhibit strong agglomeration capabilities. To resist the siphoning effects of a Jacobs-style city, a neighbouring city may weaken the tendency to develop towards that city. It is believed that local industrial competition contributes to the emergence of urban innovation and industrial transformation and upgrading. However, competition that is too fierce indicates that the scale of an enterprise is generally small, and the demand for extensive co-operation is lower, which cannot promote intercity relations. In addition, the decentralization of the Chinese government's economic decision-making power and fiscal decentralization has led to fierce competition between local governments. Fierce competition within cities will intensify competition between cities and weaken intercity relations.

Although the adjusted-R square slightly decreased when applying industrial structure similarities and interaction terms, the influence of relevant independent variables remains significant. The positive impact of industrial structure similarities on intercity relations is in line with common knowledge. The presence of similar industrial structures contributes to industrial concentration and clustering and benefits mechanisms of sharing, matching and learning in further promoting intercity relations. In models (3) and (5), interactions between industrial specialization and industrial competition (PS*Comp) show a significantly negative impact on intercity relations. PS*Comp denotes that competition will weaken the positive effect of industrial specialization on intercity relations. Larger enterprises usually exhibit stronger external connections and knowledge spillover capabilities. However, according to Marshall's theory, when intra-city competition is more intense, the scale of an enterprise will be limited, which will weaken the positive impact of industrial specialization on intercity relations. In model (4), interactions between industrial diversification and industrial competition (PD*Comp) are not significant.

6. Discussion

6.1. Participation of cities in regional spatial integration and coordinated development

The results show that intercity relations are unbalanced and that promoting regional spatial integration and coordinated development is not the common objective for all cities. The observed differences in participation can be attributed to three factors: demand, motivation and ability. (1) Demand. When an industrial structure is clearly different and when industry interdependence and linkages are weak, regional spatial integration and coordinated development are less beneficial to the economic development of cities. Local governments create an insufficient internal impetus to participate in regional coordinated development and are more willing to invest considerable costs and efforts in local development for political gains. (2) Motivation. According to the cumulative causation model, regional primary cities with a first-mover advantage exhibit stronger agglomeration capability. Some ordinary cities exhibit weak fixity and agglomeration abilities. Due to the influence of "siphoning effects", they cannot secure the benefits and advantages of regional coordinated development and instead become

hinterland of cities with stronger agglomeration capability to absorb various resources. They will then lose their power and ability to seek stronger development due to a backwash effect (Mankiw, Romer, & Weil, 1992). This series of reasons will greatly weaken the motivations of some cities to promote regional coordinated development. (3) Ability. The development of cities depends on the quantity and quality of various development factors and on the ability to acquire them. Although the rapid development of infrastructure and transportation has shortened spatial distances between cities, spatial segmentation caused by natural geographical elements has caused the effects of agglomeration and diffusion to be exerted insufficiently, which is not conducive to clusters and scale advantages. More importantly, China's territorial administrative divisions follow a clear hierarchical structure. Differences in administrative rank shape policymaking power, the ability to attract investments from the central government and from abroad, and the role of local governments. It should be emphasized that these three factors do not exist independently and often interact to influence the participation of cities in regional coordinated development.

6.2. Redefining the role of government and ensuring rational planning

Existing research shows that Chinese regional development is influenced by two different regional-scale mechanisms: top-down state-mandated processes and bottom-up processes initiated by local governments (Li & Wu, 2017). Although the above findings support this conclusion, the bottom-up process initiated by local governments has only emerged in certain areas of the YRD. The top-down state-mandated process has been more prominent in the YREB. There is no doubt that politics and policies will play a pivotal role in shaping urban agglomeration and in achieving regional spatial integration and coordinated development. However, basic institutions involved and strategies adopted are not satisfactory when creating a new sub-national state place. First, horizontal linkages between cities are weak within hierarchical administrative structures in the midst of political centralization and fiscal decentralization, cadre promotion, rent-seeking and other factors exacerbating horizontal linkages and intensified competition among local governments. Second, the developing and implementation of various development plans led by governments sometimes do not conform to objective facts or laws. Development plans have become new channels through which some local governments have pursued local interests rather than participating in regional coordinated development. For example, when delineating the scope of a metropolitan area, intercity relations and tendencies of urban system self-organization have been considered less, and priority has been given to administrative boundaries such as the Shanghai and Suxichang metropolitan areas and the Nanjing and Hefei metropolitan areas. Some regional plans are ambitious, vague and likely difficult to achieve, such as the MRYR plan. To cope with this series of development dilemmas, political institutions and government roles must be adjusted. The central government has recently replaced the business tax with a value-added tax and has merged national and local taxation administration in addition to making other adjustments, laying the foundations for the reconstruction of fiscal relationships between the central and local governments. These policies would undoubtedly strengthen the central government's guiding and coordinating role in regional development and conflicts and address issues that hinder regional coordinated development. In addition, regional coordinated development involves the restructuring of the entire social system, and it is also necessary to shift the government's role from a focus on management to a focus on governance while encouraging social participation. Through compulsory management schemes such as those of natural resource protection, the government must build sound coordination mechanisms to incentivize diversified participants to engage in regional development rather than using direct intervention.

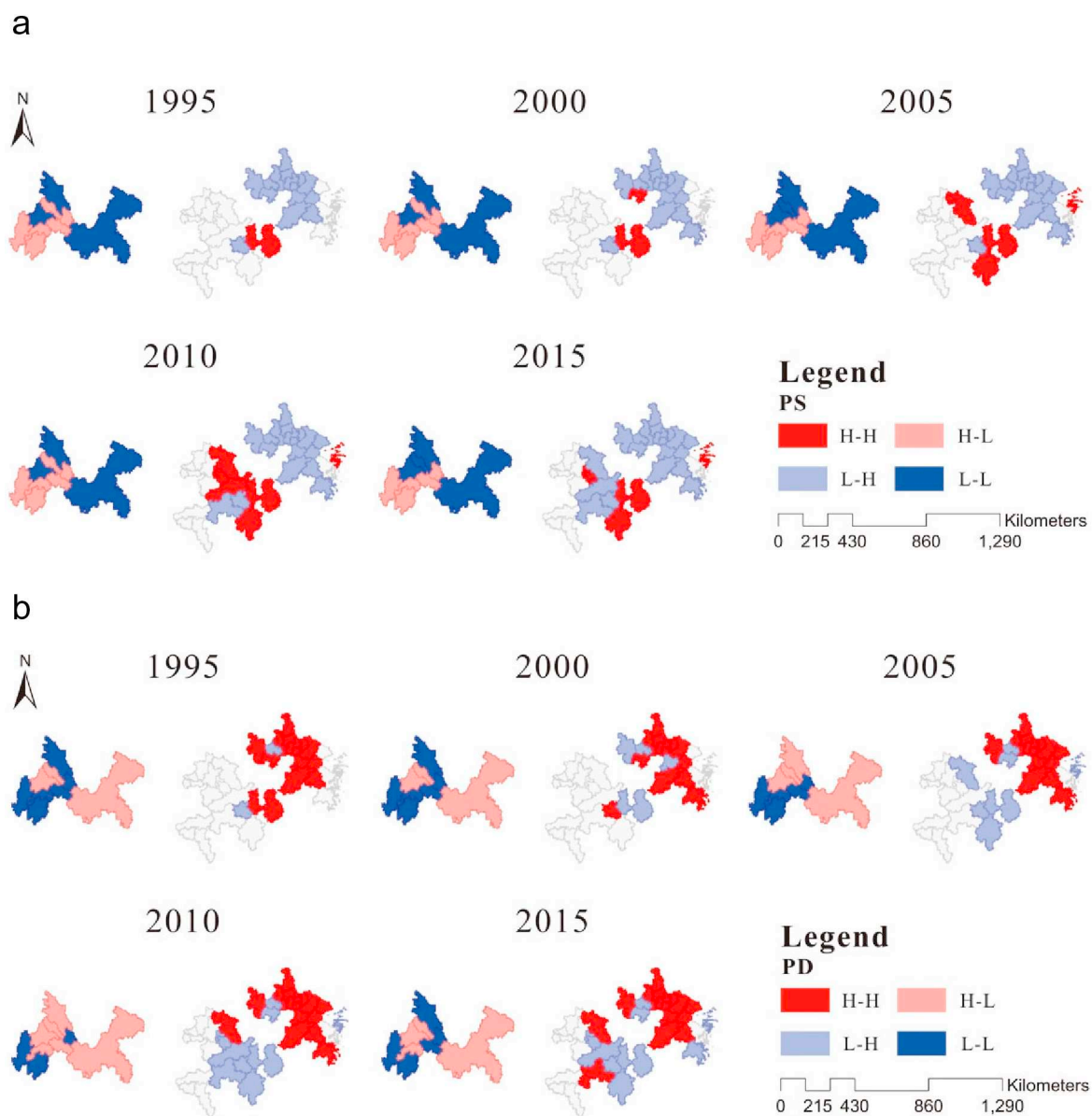


Fig. 6. (a) LISA cluster maps of relationships between industrial specialization and urban form. (b) LISA cluster map of relationships between industrial diversification and urban form.

Table 3

Panel regression results for the relationship between intercity relations and industrial characteristics.

	(1)	(2)	(3)	(4)	(5)
Ln(PS)	-0.0865	-0.0751	0.0124	-0.0757	0.2512
Ln(PD)	-0.3594***	-0.4706***	-0.6121***	-0.4768**	-0.0369
Ln(Comp)	-0.0390*	-0.0594***	0.0887	-0.0628	0.5793*
Ln(ISS)		0.3243	0.6196**	0.3287	0.7205*
Ln(PS)*Ln(Comp)			-0.1507*		-0.3097***
Ln(PD)*Ln(Comp)				0.0040	-0.4251*
Constant	-2.3441***	-2.1465***	-2.0217	-2.1394***	-2.6723***
R-squared	0.9861	0.9732	0.9628	0.9731	0.9547
Adjusted R-squared	0.9825	0.9661	0.9528	0.9658	0.9423
F-statistic	273.5205	137.3916	96.2642	134.4139	77.1206
Prob(F-statistic)	0.0000	0.0000	0.0000	0.0000	0.0000
Hausman Test	10.3271(0.0160)	18.6871(0.0009)	15.7452(0.0076)	19.0681(0.0019)	14.3003(0.0265)

Note: *** Significance at 1%; ** Significance at 5%; * Significance at 10%.

7. Conclusion

As a driver of regional development and prosperity, urban agglomeration, a new form of spatial governance, is emerging and rapidly proliferating across the globe and has attracted interests from urban geographers, planners and policymakers. This paper provides a new perspective for research on the spatial integration of urban agglomerations and identifies relationship between spatial integration and industrial development. Drawing on nighttime light data, we measure and analyse urban forms and intercity relations observed during the spatial evolution of urban agglomerations, and six types of changes in intercity relations observed in four consecutive periods are further identified and explored. Our results confirm that the spatial forms of most cities underwent apparent changes during the study period closely related to city ranks, locations and GDP increments. In addition, the spatial evolution of the three urban agglomeration patterns examined reveal clear regional differences in the YREB and that intercity relations and spatial systems have not yet fully established or matured in the central and western urban agglomerations. From a dataset on Chinese industrial enterprises, the calculated bivariate Moran's I value shows a significantly negative (positive) correlation between industrial specialization (diversification) and urban form, and LISA cluster maps show the presence of different spatial clustering patterns in the three agglomerations. The panel regression provides mixed results on the effects of characteristics of industrial development on intercity relations. Stronger similarities in industrial structures could improve intercity relations and perhaps through the benefits of more economies of scale and agglomeration economies. Industrial diversification is negatively correlated with intercity relations. To resist backwash and siphoning effects brought about by the Jacobs-style city, cities do not tend to strengthen their relations. In addition, fierce competition does not benefit intercity relations. Although all sectors of society actively embrace urban agglomeration and advocate for regional spatial integration and coordinated development, conflicts between development interests and goals among cities are evident, and barriers to urban agglomeration development planning and design remain.

These findings contribute a body of evidence on the spatial integration of urban agglomeration and on relationships between spatial integration and industrial development and specifically in reference to the Chinese context. They illustrate the current state of urban agglomeration and its internal intercity relations and can guide future studies on regional spatial integration and coordinated development. However, our analysis presents the following limitations, which in turn reveal avenues for future research. First, the study mainly analyses urban agglomeration from a form-dominant perspective. Function-dominant research focuses and methods in terms of economic connections, population and information flows, transportation and other factors should be improved through urban agglomeration studies through the use of multisource flow data. Comparative research should also be conducted with a focus on two research paradigms. Second, this study only focuses on the prefecture scale. Future studies should conduct examine specific industrial sectors and the spatial evolution of peri-urban patterns especially in certain megacities to further understanding of regional spatial integration and coordinated development. Third, NPP/VIIRS data have several confounding factors, such as seasonal changes (Levin, 2017) and satellite angular effect (Li et al., 2019). The research results may be improved if more comprehensive factors are considered.

CRedit authorship contribution statement

Yaolin Liu: Supervision, Funding acquisition. **Xianghui Zhang:** Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. **Xingyu Pan:** Resources, Formal analysis. **Xiuxin Ma:** Visualization. **Mingyang Tang:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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